

Day 5 Solutions:

1. Three Divisors

A number has exactly three divisors if and only if it is a perfect square of a prime number. For example, 4 (divisors: 1, 2, 4) and 9 (divisors: 1, 3, 9) follow this rule, while a prime number like 5 has only two divisors (1, 5). The provided code takes a direct simulation approach by counting every integer from 1 up to n that divides n without a remainder. If the final count is exactly 3, the condition is met.

1. Initialize an integer variable divisor to 0 to keep track of the total number of divisors found.
2. Run a for loop from $i = 2$ up to and including $n/2$.
3. Inside the loop, check if n is divisible by i using the modulo operator ($n \% i == 0$).
 - If the condition is true, increment the divisor count.
4. After the loop completes, use a ternary operator to check if divisor is equal to 3. (1 and n is divisors for all numbers by default)
 - Return true if the count is exactly 3; otherwise, return false.

Code:

```
class Solution:

    def isThree(self, n: int) -> bool:

        c=0

        for i in range(2,n//2+1,1):

            if n%i==0:

                c+=1

        return True if c==3 else False
```

Complexity:

Time Complexity: $O(n)$

→ We traverse the array once.

Space Complexity: $O(1)$

→ Only two variables are used.